Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/19/2014
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	06/08/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	ACES (Atomic Clock Ensemble in Space) is an ESA project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We propose to assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify		

	workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our cavity work produced an improved microwave cavity design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British and South Korean National Laboratories, National Physical Laboratory (NPL), and KRISS. The new cavity design allows significantly reduced clock errors due to Doppler shifts.
Task Progress:	Since our last annual report, we have published a paper in Physical Review Letters, “Scattering of Cold Atom Coherences by Hot Atoms: Frequency Shifts from Background Gas Collisions,” Kurt Gibble, Phys. Rev. Lett. 110, 180802 (2013). The paper treats the largest error common to some of the most accurate primary atomic clocks (Paris and NPL), the frequency shift due to background gas collisions. All reports to BIPM have used a room-temperature measurement for laser-cooled fountain clocks. The paper shows that the room-temperature result does not apply, the shift for cold atoms is less, and a new way to evaluate the uncertainty by bounding the loss of Ramsey fringe contrast due to background gas collisions. This result has already been applied to the cesium clocks in the UK and Paris, to optical lattice clocks at JILA and PTB, and will be useful for the ACES/PHARAO project. We have constructed dense 3D finite-element models of the PHARAO cavity and feed structure and solved for the standing-wave and loss fields. We have used the calculated field to evaluate the microwave lensing frequency shift of the PHARAO clock. This is currently the largest systematic in the PHARAO error budget. With this evaluation, it becomes negligible. We have extensively modeled the coupling of the feed and clock cavities of the structure, raising questions about the dimension and construction of the cavity. Work on this is ongoing, in collaboration with the team in Paris. During the past year, we reduced the memory required for our finite-element calculations by a factor of two. This is particularly important since our 1TB of RAM gives a mesh density that is not overly dense. We have solved for the loss field and have calculated clock frequency shifts due to the distributed cavity phase shifts that result from the loss fields. We have also implemented calculations for resistance inhomogeneities to explore potential accuracy limitations of PHARAO. Work on the analysis of these frequency shifts is ongoing. We have made precise measurements of scattering phase shifts of ultra-cold scattering of cesium atoms. Work included reducing a number of systematic errors, including magnetic field gradients, inelastic collisions, and cold collision frequency shifts. An extensive mathematical model of the experiment was developed to analyze the systematic uncertainties. Preliminary indications are encouraging and work is continuing. These measurements are expected to provide precise information needed to predict the ultracold collision frequency shift of PHARAO. A number of talks were presented at conferences during the past year. Three talks were given at the 2013 Joint meeting of the European Frequency and Time Forum and the International Frequency Control Symposium in Prague, an invited talk at at Quantum to Cosmos 6 in Nice France and at the 2013 American Society for Gravitational and Space Research annual meeting in Orlando, and at a summer school in Germany.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Abstracts for Journals and Proceedings	Peterman P, Gibble K, Laurent P, Salomon C. "Accuracy Evaluation of the ACES/PHARAO Laser-cooled Atomic Space Clock." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. K. Gibble presented a talk. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 48. See https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Articles in Peer-reviewed Journals	Gibble K. "Scattering of cold-atom coherences by hot atoms: frequency shifts from background-gas collisions." Phys Rev Lett. 2013 May 3;110(18):180802. PubMed PMID: 23683186 ; http://dx.doi.org/10.1103/PhysRevLett.110.180802 , May-2013
Papers from Meeting Proceedings	Gibble K. "Scattering of cold atom coherences by hot atoms: Background gas collision shifts of primary fountain clocks." 2013 Joint European Frequency and Time Forum & International Frequency Control Symposium, Prague, Czech Republic, July 21-25, 2013. In: 2013 Joint European Frequency and Time Forum & International Frequency Control Symposium (EFTF/IFC) Conference Proceedings, 2013. p. 46-47. http://dx.doi.org/10.1109/EFTF-IFC.2013.6702259 , Jul-2013